

KE-04: Strict interval occupancy across block Lanczos iterations

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Status of this manuscript: Independently checked resolution (Codex-agent review).

Outcome: Affirmative resolution.

Prepared: 11 September 2026.

Target: [Repository entry KE-04](#), snapshot b412319.

The original proof draft was generated in a ChatGPT conversation. A separate Codex agent independently checked the complete proof against the exact repository target on 11 September 2026; its [detailed review](#) records **PASS**. The theorem and proof text below are unchanged from the reviewed submission. This verification is an independent agent review, not external human peer review or a formal proof certificate.

Theorem KE-04: proposed strict interval occupancy

Let $A \in \mathbb{R}^{n \times n}$ be symmetric and let $V \in \mathbb{R}^{n \times p}$ have full column rank. Define

$$\mathcal{K}_\ell = \text{range}[V, AV, \dots, A^{\ell-1}V].$$

Assume $\dim \mathcal{K}_s = sp$ and therefore full block dimension at all preceding iterations. Let P_ℓ be the orthogonal projector onto $\mathcal{K}_\ell$, and let $H_\ell = P_\ell A|_{\mathcal{K}_\ell}$. Write its eigenvalues, with multiplicity, as

$$\theta_1^{(\ell)} \leq \dots \leq \theta_{\ell p}^{(\ell)}.$$

These are the Ritz values and are independent of the chosen orthonormal basis.

Claim. For every $1 \leq k < j \leq s$ and every $1 \leq i \leq (k-1)p$, the open interval

$$(\theta_i^{(k)}, \theta_{i+p}^{(k)})$$

contains an eigenvalue of H_j .

1. Assume a gap at the later iteration

There are no admissible indices when $k = 1$, so suppose $k \geq 2$. Put

$$a = \theta_i^{(k)}, \quad b = \theta_{i+p}^{(k)}, \quad q(t) = (t-a)(t-b).$$

Suppose that H_j has no eigenvalue in (a, b) . Since $a \leq b$,

$$q(H_j) \geq 0.$$

Let $E \subseteq \mathcal{K}_k$ be the span of an orthonormal set of eigenvectors of H_k corresponding to indices $i, \dots, i+p$. Then

$$\dim E = p+1, \quad \langle x, q(H_k)x \rangle \leq 0 \quad (x \in E).$$

The codimension of $\mathcal{K}_{k-1}$ in $\mathcal{K}_k$ is p , so $E \cap \mathcal{K}_{k-1}$ contains a nonzero vector x .

2. The quadratic forms agree on that vector

Since $x \in \mathcal{K}_{k-1}$, we have $Ax \in \mathcal{K}_k$ and

$$H_k x = H_j x = Ax.$$

Self-adjointness of the compressions gives

$$\begin{aligned}\langle x, q(H_j)x \rangle &= \|Ax\|^2 - (a+b)\langle x, Ax \rangle + ab\|x\|^2 \\ &= \langle x, q(H_k)x \rangle \leq 0.\end{aligned}$$

Combined with $q(H_j) \succeq 0$, this implies $q(H_j)x = 0$.

Moreover $A^2x \in \mathcal{K}_{k+1} \subseteq \mathcal{K}_j$, so $H_j^2x = A^2x$. Hence

$$q(A)x = 0.$$

3. Full block rank excludes this annihilation

Write

$$x = \sum_{\ell=0}^{k-2} A^\ell V c_\ell.$$

Let r be the largest index with $c_r \neq 0$; such an index exists because $x \neq 0$. The polynomial q is monic of degree two. Therefore the coefficient of $A^{r+2}V$ in the resulting block-power expansion of $q(A)x$ is exactly $c_r \neq 0$.

Since $k+1 \leq j \leq s$, the block Krylov matrix

$$[V, AV, \dots, A^k V]$$

has full column rank. As $r+2 \leq k$, the nonzero coefficient just found implies $q(A)x \neq 0$, a contradiction.

This establishes the claim for all admissible indices and iterations. If $a = b$, the same argument applies with $q(t) = (t-a)^2$ and rules out that case; thus the proof does not presuppose strictly separated endpoints or simple Ritz values. $\square$

Scope and review notes

The argument uses real symmetric A , a full-column-rank starting block, exact arithmetic, and full block dimension through iteration j . It includes eigenvalue multiplicities and the possibility of coincident proposed interval endpoints, which it rules out in the relevant range. It does not assert a finite-precision analogue.

The independent review checked the equality of the two quadratic forms on a vector in $\mathcal{K}_{k-1}$, followed by conversion to a degree-two annihilating polynomial. Full rank is required through $\mathcal{K}_{k+1}$, which is guaranteed by $k+1 \leq j \leq s$.

The packaging pass checked the repository statement and contribution rules on 11 September 2026. It did not conduct a new exhaustive literature search or establish novelty.

Source references

Alex Townsend, *Open Problems in Numerical Linear Algebra* (2026), [entry KE-04](#), repository snapshot b412319, accessed 11 September 2026.

D. Šimonová and P. Tichý, *On finite precision block Lanczos computations*, [arXiv manuscript, version 1](#), 22 July 2025, §2.1, unnumbered conjecture. The catalog identifies this as the source of its exact-arithmetic interval statement.